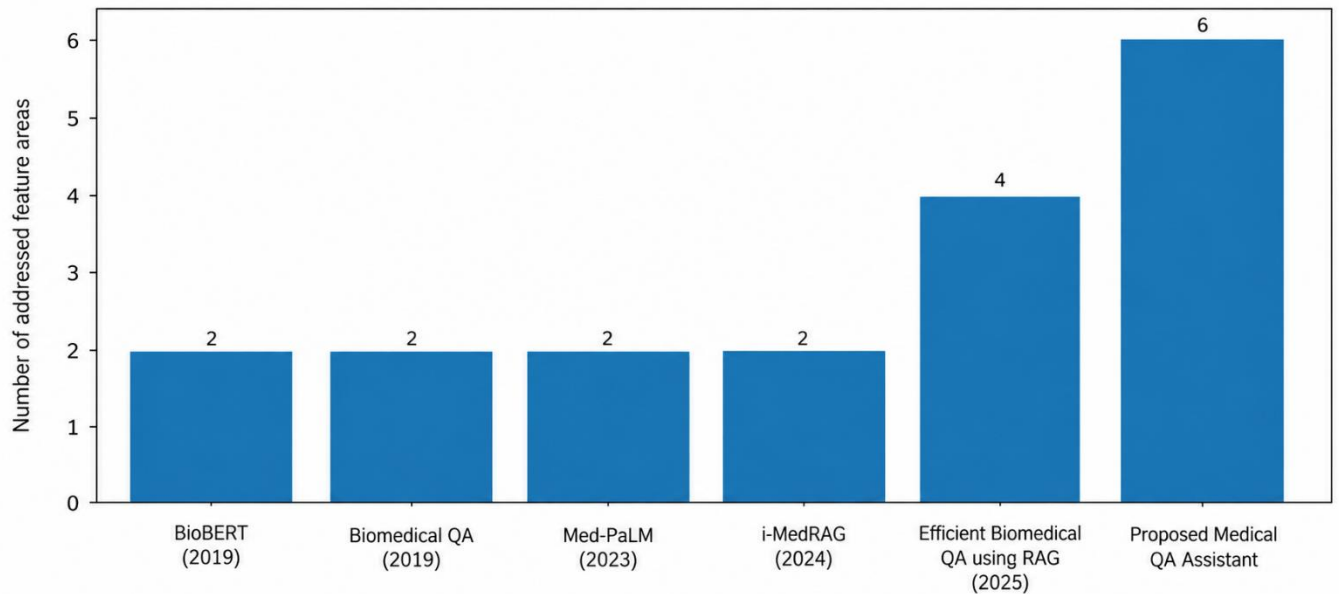




3. Comparative Analysis / Research Gap

- Existing medical question-answering approaches mainly focus on **biomedical language understanding, medical answer generation, or retrieval-based question answering**.
- BioBERT-based approaches improve biomedical text understanding, but they **do not retrieve relevant medical documents before generating answers**.
- LLM-based approaches such as Med-PaLM provide medical answer generation, but they have limitations related to **proprietary access and computational requirements**.
- RAG-based approaches improve retrieval-grounded medical QA, but some approaches have **complex architectures and higher processing time**.
- The 2025 RAG-based biomedical QA approach combines **RAG, FAISS and BioBERT**, but is mainly research-focused and is **not presented as a complete web application**.
- Therefore, a gap remains in combining **trusted medical retrieval, biomedical understanding, RAG-based generation, source support, and a complete user-oriented web application** in one framework.
- The proposed system addresses this integration gap by combining **MedQuAD, BioBERT, FAISS, RAG, LangChain and an LLM**, together with source-supported responses and interactive web features.

